12.1 Probability and Sample Spaces

The **sample space** S is the set of all possible outcomes of an experiment; an **event** E is any subset of those outcomes. For equally likely outcomes, $P(E) = \dfrac{n(E)}{n(S)}$ — the count of favorable outcomes over the total count. Two techniques make counting outcomes manageable: listing/organizing the full sample space (a grid for two dice, a tree for sequential flips), and the **complement rule** $P(E') = 1 - P(E)$, which is often far faster than counting the event directly — especially for "at least one" questions. All probability answers in this course should be exact fractions in lowest terms.

Examples

Example 1: Single die

A fair six-sided die is rolled. Find $P(\text{rolling a } 4)$.

Solution

Sample space: $S = \{1, 2, 3, 4, 5, 6\}$, so $n(S) = 6$. Only one outcome is favorable.

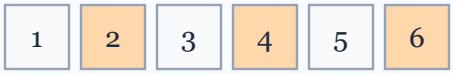
$P(4) = \dfrac{1}{6}$.

Answer: $\dfrac{1}{6}$.

Example 2: Event with multiple outcomes

A fair die is rolled. Find $P(\text{even})$.

Solution



Favorable outcomes {2,4,6} highlighted out of the full sample space {1,...,6}.

Favorable outcomes: $\{2, 4, 6\}$, so $n(E) = 3$; $n(S) = 6$.

$P(\text{even}) = \dfrac{3}{6} = \dfrac{1}{2}$ (always reduce to lowest terms).

Answer: $\dfrac{1}{2}$.

Example 3: Marbles

A bag has 5 red, 3 blue, and 2 green marbles. One is drawn at random. Find $P(\text{green})$.

Solution

$n(S) = 5 + 3 + 2 = 10$ total marbles; $n(E) = 2$ green.

$$P(\text{green}) = \frac{2}{10} = \frac{1}{5}.$$

Answer: $\frac{1}{5}$.

Example 4: Complement rule

Same bag (5 red, 3 blue, 2 green). Find $P(\text{not red})$ using the complement.

Solution

$$P(\text{red}) = \frac{5}{10} = \frac{1}{2}.$$

$$P(\text{not red}) = 1 - P(\text{red}) = 1 - \frac{1}{2} = \frac{1}{2}.$$

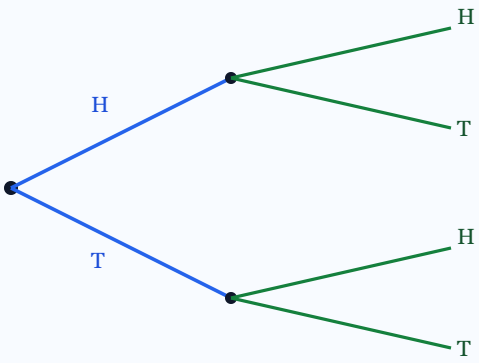
Check directly: not red means blue or green: $\frac{3+2}{10} = \frac{5}{10} = \frac{1}{2}$. ✓

Answer: $\frac{1}{2}$.

Example 5: Two coins — sample space tree

Two fair coins are flipped. Find P(exactly one head).

Solution



The tree shows all 4 equally likely outcomes: HH, HT, TH, TT.

$S = \{HH, HT, TH, TT\}$, so $n(S) = 4$. "Exactly one head" matches HT and TH: $n(E) = 2$.

$P(\text{exactly one head}) = \frac{2}{4} = \frac{1}{2}$.

Answer: $\frac{1}{2}$.

Example 6: Spinner

A spinner is numbered 1 through 8, all equally likely. Find P(greater than 5).

Solution

Favorable: $\{6, 7, 8\}$, so $n(E) = 3$; $n(S) = 8$.

$P = \frac{3}{8}$ (already in lowest terms).

Answer: $\frac{3}{8}$.

Example 7: Standard deck of cards

One card is drawn from a standard 52-card deck. Find $P(\text{heart})$.

Solution

There are 13 hearts in a deck of 52 cards.

$P(\text{heart}) = \frac{13}{52} = \frac{1}{4}$ (divide by the GCF, 13).

Answer: $\frac{1}{4}$.

Example 8: Complement with a jar

A jar has 6 yellow and 9 purple marbles. Find $P(\text{not yellow})$.

Solution

$n(S) = 6 + 9 = 15$. $P(\text{yellow}) = \frac{6}{15} = \frac{2}{5}$.

$P(\text{not yellow}) = 1 - \frac{2}{5} = \frac{3}{5}$.

Answer: $\frac{3}{5}$.

Example 9: Three coins

Three fair coins are flipped. Find $P(\text{all three match})$.

Solution

S has $2^3 = 8$ equally likely outcomes: $HHH, HHT, HTH, HTT, THH, THT, TTH, TTT$.

"All match" means HHH or TTT: $n(E) = 2$.

$P = \frac{2}{8} = \frac{1}{4}$.

Answer: $\frac{1}{4}$.

Example 10: Two dice — sample space grid

Two fair dice are rolled. Find $P(\text{sum} = 7)$.

Solution

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

All 36 equally likely ordered pairs from rolling two dice.

$n(S) = 6 \times 6 = 36$. Sum of 7: $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$ — exactly 6 outcomes.

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}.$$

Answer: $\frac{1}{6}$. *(Sum of 7 is the single most likely sum on two dice.)*

Example 11: "At least one" — complement with two dice

Two fair dice are rolled. Find $P(\text{at least one 6})$.

Solution

Counting "at least one 6" directly is messy — use the complement: "no 6 on either die."

Each die has 5 non-6 outcomes, so $n(\text{no 6}) = 5 \times 5 = 25$ out of 36.

$$P(\text{no 6}) = \frac{25}{36}, \text{ so } P(\text{at least one 6}) = 1 - \frac{25}{36} = \frac{11}{36}.$$

Answer: $\frac{11}{36}$.

NAME: _____ Geometry (FL B.E.S.T.)

Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Die

Fair die. Find $P(\text{odd})$.

Answer: **Verification:**

Problem 2: Die

Fair die. Find $P(\text{at least } 5)$.

Answer: **Verification:**

Problem 3: Marbles

Bag: 4 red, 6 blue. Find $P(\text{red})$.

Answer: **Verification:**

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Problem 4: Complement

Bag: 4 red, 6 blue. Find P(not blue).

Answer: **Verification:**

Problem 5: Coins

Two fair coins. Find P(two heads).

Answer: **Verification:**

Problem 6: Spinner

Spinner 1–10. Find P(multiple of 3).

Answer: **Verification:**

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Problem 7: Cards

Find $P(\text{ace})$ from a 52-card deck.

Answer: **Verification:**

Problem 8: Sample space

List the sample space S for one flip of a fair coin.

Answer: **Verification:**

Problem 9: Range

What is the range of possible probability values?

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 10: Certain event

Find P(die shows 1 through 6).

Answer: **Verification:**

Problem 11: Impossible event

Find P(die shows 7).

Answer: **Verification:**

Problem 12: Three coins

Three fair coins. Find P(exactly two heads).

Answer: **Verification:**

Problem 13: Complement

$P(E) = 0.35$. Find $P(E')$.

Answer: **Verification:**

Problem 14: Two dice

Two fair dice. Find $P(\text{sum} = 8)$.

Answer: **Verification:**

Problem 15: Challenge

Two fair dice are rolled. Find $P(\text{sum is a prime number})$ — remember $2, 3, 5, 7, 11$ are the possible prime sums.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

✓ Answer Key

1. $\frac{1}{2}$
2. $\frac{1}{3}$
3. $\frac{2}{5}$
4. $\frac{2}{5}$
5. $\frac{1}{4}$
6. $\frac{3}{10}$
7. $\frac{1}{13}$
8. $\{H, T\}$
9. 0 to 1, inclusive
10. 1
11. 0
12. $\frac{3}{8}$
13. 0.65
14. $\frac{5}{36}$
15. $\frac{5}{12}$

Great work — keep practising!